higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

PLUMBING
NQF LEVEL 4

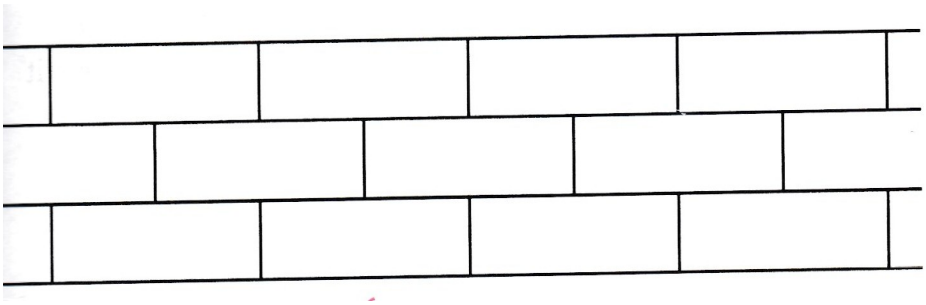
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This marking guideline consists of 6 pages.

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QUESTION 1: BASIC BUILDING WORK AND MANHOLES

1.1



Three layers	1
Correct bond	1
Neatness	1

(3)

1.2 A manhole is an inspection chamber✓ deeper than 750 mm and big enough for a person to enter the chamber to reach the drain. .✓ (2)

1.3 First lay a blinding layer of concrete. It should be at least 50 mm thick✓ and must withstand compression of 10 MPa after 28 days.✓
The foundation is cast on the blinding layer.✓ It must be at least 150 mm thick at the minimum point✓ and withstand a minimum compression of 20 MPa after 28 days.✓ (4)

1.4 A soffit is the top point of the inside open section of pipe. (1)
[10]

QUESTION 2: BELOW GROUND DRAINAGE SYSTEM

2.1 A rodding eye is an opening in a drainage installation that allows access to the drainage system. (2)

2.2
- Set out warning lights or other visible lights so that the boundaries of the excavation are clearly visible.
- Surround the excavation with reflective danger tape. (2 × 1) (2)

2.3 Builder, architect or another responsible person. (1)

2.4 2.4.1 1:40
2.4.2 1:60
2.4.3 1:90
(3 × 1) (3)

2.5
- A water level.
- Boning rods and sight rails.
- A dumpy level with a tripod and measuring staff.
- A spirit level, a straight edge, a wooden gradient template and metal or wooden pegs.
- Profile boards. (5 × 1) (5)

2.6 Bedding is the layer of sand or other material on the bottom of the trench.✓ on which the drainage pipes rest. .✓ (2)

2.7
- In the air test a special compressor is used.
- In the water test water is poured into a stack using a hose. (2 × 1) (2)

2.8
- SANS 1200LD
- The NHBRC
- SANS 10252 (3 × 1) (3)

[20]

QUESTION 3: PLUMBING MAINTENANCE AND REPAIRS

3.1 To look pretty (1)

3.2
- First close the stop valve.
- Cut out the pipe of pipe where the leak is.
- Measure the length of pipe that you have removed.
- Cut a new piece of pipe 2 mm shorter than the piece that you have removed.
- Use a special slip socket of a slightly larger diameter than the pipe to hold the old and the new pipe together.
- Use sand paper to clean the ends of all copper pipes.
- Apply flux to both the inside of the socket and outside of the pipe.
- Fit the pieces together.
- Use a blow torch to heat the pipe and solder it.
- Remove any excess flux after the joint has been soldered. (Any 9 × 1) (9)

3.3 To control the temperature in the geyser. (1)

3.4
- Turn off the main water supply to the geyser.
- Loosen and remove the faulty valve by unscrewing it by hand.
- Screw the new valve in by hand taking care not to damage the thread.
- Turn on the water supply to the geyser. (4 × 1) (4)

3.5
- A plunger
- Rods
- A kinetic ram gun
- Coring and scraping
- Chemical cleaning (5)

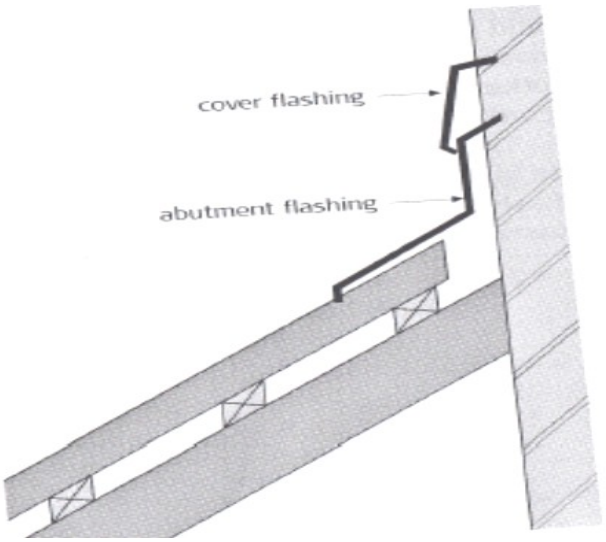
[20]

QUESTION 4: BOX GUTTERS AND CHIMNEY FLASHING

4.1 A box gutter is a rectangular gutter that is positioned inside or against the wall✓ and does not hang over the edge of the roof. ✓ (2)

4.2
- Apply bituminous paint, alkali-resistant lacquer or zinc chromate paint.
- Use a moisture-proof membrane (DPM) such as plastic film, bitumen impregnated paper, or felt. (2)

4.3



Drawing	2
Label	2

(4)

4.4
- Aluminium and aluminium alloys
- Milled lead (Any 1 × 1) (1)

4.5
- Measure the chimney and its surrounds carefully.
- Draw the outlines and fold lines on a large piece of paper.
- Draw the step flashing.
- Cut out the apron flashing and the back gutter.
- Cut and fold the step flashing.
- Fit the flashing to the chimney wall. (6)

[15]

QUESTION 5: INSTALL A SOLAR WATER HEATER

5.1 A solar collector is the part or component of a solar heating system that absorbs the sun's energy .✓ and converts it to heat. .✓ (2)

5.2 Radiant energy is sent out✓ in the form of waves. ✓ (2)

5.3 5.3.1 energy from the sun
5.3.2 feel every day
5.3.3 makes us hot
5.3.4 radiant energy
5.3.5 life-giving energy
5.3.6 source of energy
5.3.7 run out
5.3.8 harmful by-products

(8 × 1) (8)

5.4 Because the back-up system will provide hot water on cloudy days✓ and during periods of increased demand. ✓ (2)

5.5 Standby heat loss refers to the heat energy wasted by keeping the water in the storage tank hot✓ while there is no demand for it. ✓ (2)

5.6 The fins (or fin coating). (1)

5.7 The position✓ of the collector. ✓ (2)

5.8
- Make as few roof penetrations as possible.
- Locate the rafters to which you will be attaching the collectors.
- Seal roof penetrations with silicon sealant. (3)

5.9 A demand water heater is a heater in which the water is heated✓ at the time of demand. ✓ (2)

5.10 An engineer. (1)

[25]

QUESTION 6: BASIC BUSINESS CONCEPTS

6.1 Business plan (1)

6.2
- How many other businesses offer the same service in the area?✓
  - Are they good?✓
  - Can I do a better job?✓
- Are they very busy?✓
  - Will there be enough work for all of us?✓
  - Will we be able to take some work away from them?✓ (Any 2 × 1) (2)

6.3 The work that needs to be done
The deadline for the completion of the work
The total cost of the work
When payment must be made
The amount or % that must be paid (any suitable three) (3)

6.4
- The bank does not charge the receiver a large fee for money paid into the account.
- It is a very safe method of payment.
- The bank has a system that helps one check payments received.
- The money is usually available within 48 hours. (4)

[10]

TOTAL: 100

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